

113A Introduction to Computer and Computer Science

Homework Assignment #8

Due: 11/18 12:00:00

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: A Dice Game (50%)

You have already created a “**Dice**” and “**Person**” object in the [homework#7](#). This time, we are going to use those two objects to create a **game**. Here are the rules of this **game**:

1. At the beginning of this game, 2 players have the same amount of dices. The dice type of these 2 players are in the same.
2. There is a banned list with at least one integer in the range of $[1, D_{max}]$, where D_{max} represents the maximum outcome of dice.
3. Each round, every player will roll their dices and then remove the corresponding dices whose point is equal to the element in the banned list.
4. If the dice box of the player is empty, she/he is eliminated from this game.
5. Repeat step 3 and 4 until only one person left.

Here is the sample code:

```
def solution1(config):  
    numDices = config.num  
    diceType = config.dtype  
    ban = config.ban  
    printInfo = config.print  
  
    p1 = Person(config.name1, numDices=numDices, diceType=diceType)  
    p2 = Person(config.name2, numDices=numDices, diceType=diceType)  
  
    ???
```

If everything goes well, you might have these results:

```
(nycudopcs) \113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Hsiao Bo-Bo"
--name2 "Wei Tung-Tung" --num 6 --dtype 12 --q 1 --print
Problem 1:
Hsiao Bo-Bo, State of dicebox: [11, 4, 12, 7, 3, 8]
Wei Tung-Tung, State of dicebox: [11, 5, 6, 10, 8, 2]
Ban list: [8, 6, 1, 4, 5]
=====
Round 1:
Hsiao Bo-Bo, State of dicebox: [10, 5, 5, 7, 9, 5]
Wei Tung-Tung, State of dicebox: [8, 3, 2, 12, 8, 5]
-----
Hsiao Bo-Bo, State of dicebox: [10, 7, 9]
Wei Tung-Tung, State of dicebox: [3, 2, 12]
End of round 1.
=====
Round 2:
Hsiao Bo-Bo, State of dicebox: [8, 1, 5]
Wei Tung-Tung, State of dicebox: [11, 6, 5]
-----
Hsiao Bo-Bo, State of dicebox: []
Wei Tung-Tung, State of dicebox: [11]
End of round 2.
=====
Wei Tung-Tung wins!
```

```
(nycudopcs) \113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Hsiao Bo-Bo"
--name2 "Wei Tung-Tung" --num 6 --dtype 12 --q 1 --print --ban 9 4 8 7
Problem 1:
Hsiao Bo-Bo, State of dicebox: [8, 1, 5, 7, 10, 5]
Wei Tung-Tung, State of dicebox: [11, 4, 6, 4, 4, 7]
Ban list: [9, 4, 8, 7]
=====
Round 1:
Hsiao Bo-Bo, State of dicebox: [11, 7, 9, 2, 10, 9]
Wei Tung-Tung, State of dicebox: [11, 3, 11, 3, 9, 4]
-----
Hsiao Bo-Bo, State of dicebox: [11, 2, 10]
Wei Tung-Tung, State of dicebox: [11, 3, 11, 3]
End of round 1.
=====
Round 2:
Hsiao Bo-Bo, State of dicebox: [4, 4, 9]
Wei Tung-Tung, State of dicebox: [10, 10, 1, 7]
-----
Hsiao Bo-Bo, State of dicebox: []
Wei Tung-Tung, State of dicebox: [10, 10, 1]
End of round 2.
=====
Wei Tung-Tung wins!
```

Problem #2: Statistics of N Games (50%)

So far, we have created enough functions and objects to do the next experiment, the statistics of N games. Please write a function called “**gameStats**” which aims to return the corresponding result of N games. For example, it should show this message:

```
(nycudopcs) \113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Lee C-C"
--name2 "Wang Fly-Fly" --num 6 --dtype 12 --q 2
Problem 2
Stats of 1000 games:
Lee C-C wins: 406
Wang Fly-Fly wins: 435
Draw: 159

(nycudopcs) \113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Lee C-C"
--name2 "Wang Fly-Fly" --num 6 --dtype 12 --q 2 --games 100
Problem 2
Stats of 100 games:
Lee C-C wins: 40
Wang Fly-Fly wins: 42
Draw: 18

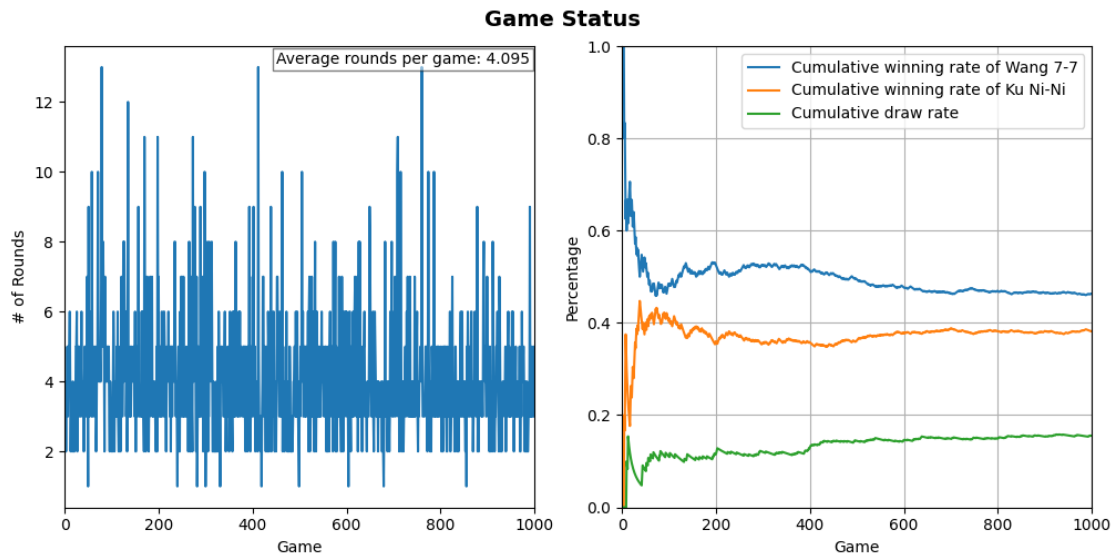
(nycudopcs) \113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Lee C-C"
--name2 "Wang Fly-Fly" --num 6 --dtype 47 --q 2 --games 100
Problem 2
Stats of 100 games:
Lee C-C wins: 36
Wang Fly-Fly wins: 58
Draw: 6
```

(Bonus) Problem #3: Plot the result of N Games (50%)

Please plot the result of the statistics of N games.

```
(nycudopcs) 113A\HW8\113A_hw8_0123456789>python main_hw8.py --name1 "Wang 7-7"
--name2 "Ku Ni-Ni" --num 6 --dtype 12 --q 3 --plot
Problem 3
Stats of 1000 games:
Wang 7-7 wins: 463
Ku Ni-Ni wins: 382
Draw: 155
```

Here is a template. You only need to provide the same information. Please feel free to change the style of this figure.

**FRIENDLY REMINDER**

There must be a method attribute in your “**Person**” to “**remove**” the dice.
 In this homework, you can use **NumPy**, **Matplotlib**, and **random** to help you.

Please accomplish this homework with an organized code (e.g., with main script and function script). **Your main script file name must be 'main_hw8.py.'** Here is a template for your code structure:

```
113A_hw8_0123456789
├─ obj.py           # Module 1 for hw8
├─ func.py          # Module 2 for hw8
├─ :                # Other modules if necessary
└─ main_hw8.py      # Main script of hw8
```

FRIENDLY REMINDER

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```
import argparse
from func import solution1, solution2, plotFigure

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2024 NYCUDOPCS
Homework #8")
    parser.add_argument("--q", default=1, type=int, help="Which
question? (default: 1)")
    parser.add_argument("--name1", default="Chen A-Hsuan", type=str,
help="Name of player 1 (default: 'Chen A-Hsuan')")
    parser.add_argument("--name2", default="Chen B-Young", type=str,
help="Name of player 2 (default: 'Chen B-Young')")
    ???
    parser.add_argument("--print", default=False,
action=argparse.BooleanOptionalAction, help="Print game log? (default:
False)")
    parser.add_argument("--plot", default=False,
action=argparse.BooleanOptionalAction, help="Plot result? (default:
False)")

    opt = parser.parse_args()

    if opt.q == 1:
        print("Problem 1:")
```

```
        solution1(
            opt
        )

    elif opt.q == 2:
        print("Problem 2")

        nRounds, nGameStats, gameLog = solution2(
            opt
        )

    elif opt.q == 3:
        print("Problem 3")

        nRounds, nGameStats, gameLog = solution2(
            opt
        )

        if opt.plot:
            plotFigure(nRounds=nRounds, gameLog=gameLog, config=opt)

    else:
        raise ValueError("The value of argument q should be 1, 2, or 3.")
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
""  
Created on Wed Aug 7 01:23:45 2024  
  
@author: Xi Winnie, student ID  
  
@collaborators: William Lai, his student ID  
""
```

Please save your homework as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113A_hw8_**ID**.zip

For example,

113A_hw8_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.